Bridging understandings of materials in sustainable product design education

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Abstract The following paper elaborates on the role of 'experience' in teaching materials for design in artistic design education to increasingly allow students to work and interact with issues related to sustainable design. The paper specifically focuses on two models used to articulate, comprehend and position experience in product design. The first model called 'the physical-experiential scale of material attributes' model demonstrates that material attributes are not either physical or experiential but exist and interrelate along a continuous scale. The scale includes the attribute categories: 'compositional properties', 'technical properties', 'sensorial attributes', 'associative characteristics' and 'emotional characteristics'. The second model called 'the five perspectives' model builds on a hierarchical understanding of sustainable thinking helping to navigate between different aspects of materials as part of sustainable design and provides a way to increasingly make these aspects support each other. The five perspectives are: 'raw materials and processes', 'products and use', 'service and systems', 'strategies and business models' and 'culture and experience' and. Acknowledging experiential dimensions and how they interact with physical dimensions can and should increasingly be regarded as co-drivers for sustainable design. In the paper the models are discussed and exemplified with product cases and student exercises.

Keywords *Materials for design, Design education, Experiential attributes, Sustainable product design, Experiential sustainability*

Introduction

As part of the rapid technological development, new materials get invented and commercialized, which means that the amount and range of materials product designers can choose from keep on increasing (Ashby & Johnson, 2014; Karana, Pedgley, & Rognoli, 2014; Manzini, 1989). For centuries (and even millennia) materials understanding has been based on processing and using materials with distinct properties and characteristics determined by the material family they belonged to. This was strongly influenced by the material's molecular composition and construction and information on conventional or traditional materials were somehow *transparent*.

With the present development and introduction of innovative materials and emerging technologies this transparency decreases. Whereas a nature-made material bears with it an identity from the environment it was grown or farmed in, a man-made material, or what we also refer to as an artificial or a synthetic material, does not possess the same kind of identity (Manzini, 1989; Vannini, 2009). Man-made materials, in product design often plastics made of nonrenewable resources from crude oil, obtain their properties from chemical processes and further manufacturing processes, where properties and appearances can be modified and altered in multiple steps.

In addition to newly developed 'smart' materials and novel technologies with properties and characteristics unlike materials previously known, in recent years materials that possess properties and characteristics of both nature- and man-made materials are increasingly becoming part of the market. These include materials such as: Tencel® from Lenzing, a textile fiber made of raw materials from birch and eucalyptus trees that have been synthetically processed and ArboForm® from Tecnaro, a so-called woodplastic composed of cellulosic fibers from wood embedded in lignin from wood that can be thermo molded in different ways, for example with injection molding. In that sense, these materials are hybrids with compositional properties and characteristics determined by the raw material they are composed of and with constructional appearance determined by the processes applied.

Sustainable development and design education

In the infancy of the 'modern' understanding of sustainable development in the early 1970s (Papanek, 1971; Schumacher, 1974) to Our Common Future report in the end of the 1980s (United Nations, 1987) main emphasis was on ecological issues relating to unsustainable consumption on nonrenewable resources, pollution caused by irresponsible production process control and large amounts of waste. Today sustainable development is much more and the value system that defines and shapes the

concept has shifted from being mainly objective to increasingly acknowledging subjective dimensions. It has shifted from perceiving sustainable initiatives as primarily singular and non-interactive entities to embracing large and complex systems with constant interactions between human and non-human actors (Bhamra & Lofthouse, 2007; Keitsch, 2015; McLennan, 2004; Vezzoli & Manzini, 2010). The broadening of the scope of sustainable development means that more disciplines take part in a collection of mindsets, methodologies and traditions that both strengthen and dilute the concept.

Sustainable development or more specifically sustainable design is an important and inevitable part of design profession. In these years most educational design institutions have or introduce courses in aspects of sustainable design and students are increasingly required to relate to sustainable issues in their work. It is a necessary step considering the challenges the society faces today, but it also risks leaving 'traditional' subjects such as materials, aesthetics and geometry in the periphery of the curriculum.

Consequently the paper is based on the research question: *"How can design education embrace the rapidly developing field of materials and simultaneously prepare students within the frames of sustainable design in everything it can mean and be?"*.

To answer this question, in this paper two models are introduced that both highlight the role of materials 'experience'. Experience is here understood as the mental recognition of a material or a physical product (made of materials) by a person after (s)he has interacted with the material or product (Dewey, 1938; Hekkert and Schifferstein, 2008; Karana et al, 2014).

The two models are based on empirical studies in an artistic design educational learning environment and have been developed to bridge theoretical gaps in the curriculum and to embrace a holistic materials teaching methodology specifically used for teaching fashion, textiles and industrial design students at Design School Kolding, Denmark, but with focus on being applicable in other design courses, artistically as well technically oriented. Here the models are exemplified with student exercises conducted in materials introductory courses and materials and sustainability courses on 2nd semester and in on 4th semester for fashion and textiles design students at Design School Kolding (spring 2014 and 2015) and in a materials course on 6th semester for sustainable design engineering students at Aalborg University, Copenhagen (spring 2016).

Model 1: Bridging physical and experiential material attributes

The breakdown of the transparency of meanings of materials calls for alternative ways to understand materials and to teach materials in design education and the first model is used to bridge physical and experiential meanings of materials.

Very simplistically two approaches to materials exist: natural sciences and engineering disciplines focus on physical aspects of materials such as mechanical, chemical and thermal properties, while human and social scientific disciplines focus on social or *experiential* aspects of materials such as, how users interact with and relate to materials (see for example Ashby & Johnson, 2014; Hasling, 2015; Johnson, Lenau, & Ashby, 2003; Karana, Barati, Rognoli, & van der Laan, 2015; Karana, Hekkert, & Kandachar, 2009; Rognoli & Levi, 2004). Design education is influenced by all of the above disciplines; designers consider technical material properties, but just as much, designers use their perception and their cognitive experience to create material meanings. In design education it is therefore essential to provide materials teaching that encourages students to consider materials and create personal material understandings based on multiple variables within technical and experiential material domains.

To do so it is necessary to consider the value systems materials take part of. A value system can be described as the collection of values human beings and communities have developed through cultural affiliation, profession, experiences and etc. (Schön, 1983; Vickers, 1968). In design engineering practice values are predominantly of technical nature such as quality, cost, (production) efficiency and time (Olesen, 1992), while values in artistic design practice are dominated by the experience and how associations and emotions are created (Schifferstein & Hekkert, 2008). Thus only considering physical properties could mean using materials in a product that are not socially suited for the product's purpose and only considering experiential characteristics produces materials that do not follow functional requirements on for example production and durability. Consequently in a material value system both technical and experiential attributes should be considered equally (Hasling, 2015) in order not to further strengthen a dichotomist approach to the materials field (Vannini, 2009).

The physical-experiential scale of material attributes

In literature on materials for design practice a plethora of taxonomies to categorize material attributes are used (Ashby & Johnson, 2014; Bang, 2010; Karana, 2009; Rognoli, 2004; van Kesteren, 2008). Based on these, a continuous span ranging from physical (objective) attributes in one extreme to experiential (subjective) attributes to the other extreme has been proposed in a model (Hasling, 2015; Hasling and Bang, In press). The 'physical-experiential scale of material attributes' model is used to illustrate, how physical properties and experiential characteristics are linked and interact. Therefore the model can be used to create attention towards specific attributes and how they interact and can be translated. The model includes five categories of attributes being: 'compositional properties', 'technical properties', 'sensorial attributes', 'associative characteristics' and 'emotional characteristics'. A short description of the attributes are given here:

— *'Compositional properties' are physical attributes*

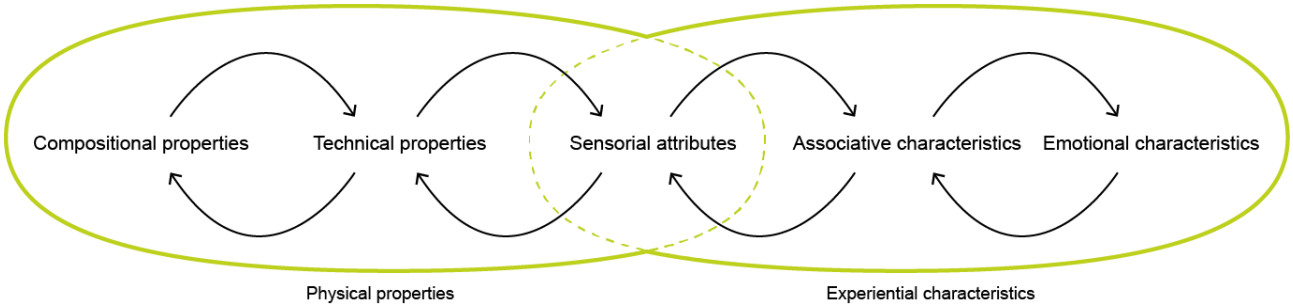


Figure 1. The continuous span of material attributes from physical properties to experiential characteristics.

determined by materials' microstructural composition,
- *'Technical properties' are physical attributes determined or influenced by processes and manufacturing methods applied,*
- *'Sensorial attributes' are physical stimuli caused by interaction between a person and one or more materials,*
- *'Associative characteristics' are the attributes related to the associations that are created based on material interaction combined with previous experiences,*
- *'Emotional characteristics' are attributes related to the emotions that are provoked by material associations.*

In Figure 1 the model shows how 'compositional properties' can be related to 'technical properties' that again can be related to 'sensorial attributes' and so on. These relations go both ways, which is indicated with arrows. Two overall categories are applied being physical properties and experiential characteristics. Physical properties include 'compositional properties', 'technical properties' and 'sensorial attributes' while experiential characteristics include 'sensorial attributes', 'associative characteristics' and 'emotional characteristics'. This leaves sensorial attributes as bridge builders, boundary objects, linking different understandings of materials from respectively physical and social worlds.

Professional designers with practical materials experience tacitly link different kinds of attributes as part of their practice and even though most people in everyday life do not take notion, linking categories of attributes are also part of how products are generally assessed. A fabric can for example be highly appreciated (emotional and/or associative characteristic) because it is soft and flexible (sensorial attributes), which is due to the construction of the fabric (technical property) and/or the choice of fiber(s) (compositional property). Similarly people's negative affections towards plastics or positive affections towards wood are associated to previous interactions based on sensorial stimuli influenced by the materials' technical and compositional properties.

The continuous scale of attributes can be used to describe and in some cases forecast, how future users will respond to a material, a technology or a product. Here the Nike Flyknit sneaker can be used as an example. The Nike Flyknit sneaker has been developed



Figure 2. The fully fashion knitted sock and the assembled shoe of a Nike Flyknit Racer (sources: www.nikeblog.com and www.dezeen.com). The inlaid fibers provide support and strength to specific parts of the shoe such as over the instep and around the ankle.

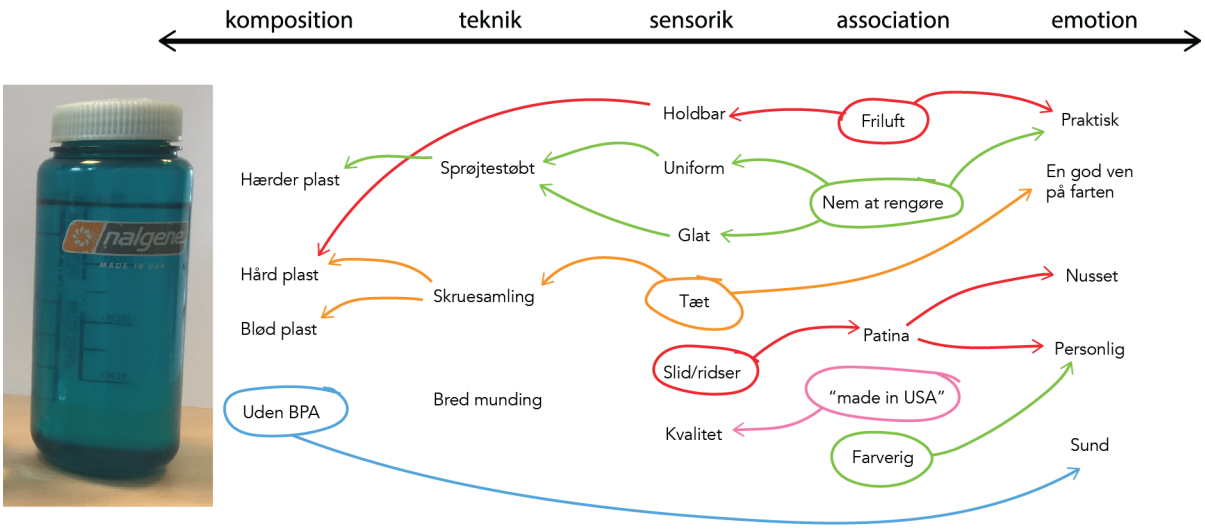


Figure 3. An example of a 'network of attributes' exercise made by a group of students in a 6th semester materials course from sustainable design engineering with focus on a plastic water bottle.

as a combination of new materials and technologies that both functionally and aesthetically have revolutionized the way sneakers appear. A newly developed polyester yarn and an injection moldable ethylene vinyl acetate (compositional properties) were combined with a knitting technology that allowed a 'sock' to be manufactured in fully fashion knit and an injection molded sole (technical properties) that made it possible to develop a sneaker with fewer parts (and thus less expensive manufacturing), lower weight, higher flexibility and customized support. The low weight and high flexibility provide a high level of comfort and the shoe is easy to maintain and can be used in different environments because of the properties of the fiber polyester (sensorial attributes). The high comfort and low maintenance makes it subject to many different associations based on its use, such as running, wet feet or holidays with warm and swollen feet (associative characteristics). These create emotional attachments based on good or bad experiences with using or associations to its such as well being, chills or feeling relaxed (emotional characteristics).

Bridging physical and experiential material attributes in materials teaching

Studies indicate that often there is a gap between the competences and knowledge of materials lecturers and the interests of design students (De Nardo & Levi, 2014). Design students' skills and interests, especially in the beginning of their studies, predominantly focus on materials applied in products, while lecturers, often with a technical background, are more competent in materials science and technologies. Consequently materials courses often become technically oriented resulting and somehow disintegrated components in the curriculum. As the model in Figure 1 demonstrates, physical properties and experiential characteristics are two sides of the same coin, and therefore materials should be introduced to students as physical and social entities equally from the beginning.

In materials teaching different approaches can be taken to encourage students to embrace and bridge physical and experiential materials attributes. In the following three examples of exercises are described.

A 'network of attributes' exercise is a very straightforward way to untangle students' mental structures, based on a specific attribute. Students are assigned to graphically sketch relationship networks incorporating subjective and objective attributes. For example can 'happiness' be associated with 'grandparents', 'pets' or 'holidays'. These can again be related to senses, objects and artefacts that are embodiments of raw materials and processes. In Figure 3 an example of a 'network of attributes' exercise is demonstrated. The network was made by a group of students in a 6th semester materials course from sustainable design engineering with focus on a plastic water bottle. Here students for example link the attributes 'hard plastic' (composition), 'durable' (sensorial), 'outdoor life' (association) and 'practical' (emotional) (upper red arrows), 'without PBA' (composition) and 'healthy' (emotional) (blue arrows) and 'wear/scratches' (sensorial), 'patina' (association) and 'shabby' and 'personal' (emotional) (lower red arrows). After the exercise the students applied the model as a frame for discussing values of products with actors in a project on student housing, which helped to provide an overview and create links between related aspects.

The 'expressive-sensorial scales' (Rognoli, 2004), the 'tripod approach' (Bang, 2010) and the 'comparative scales of material attributes' model (Hasling, 2015; Hasling & Bang, In press) are all inspired by the Bauhaus School's tactile charts building on sensorial experiences through the work with various materials (Moholy-Nagy, 1947). The exercises aim at creating correlation between simultaneous ways of working with materials and techniques and intuitively navigating between subjective and objective attributes. In Figure 4 examples of comparative scales of material attributes are shown. The exercise was made in a 2nd

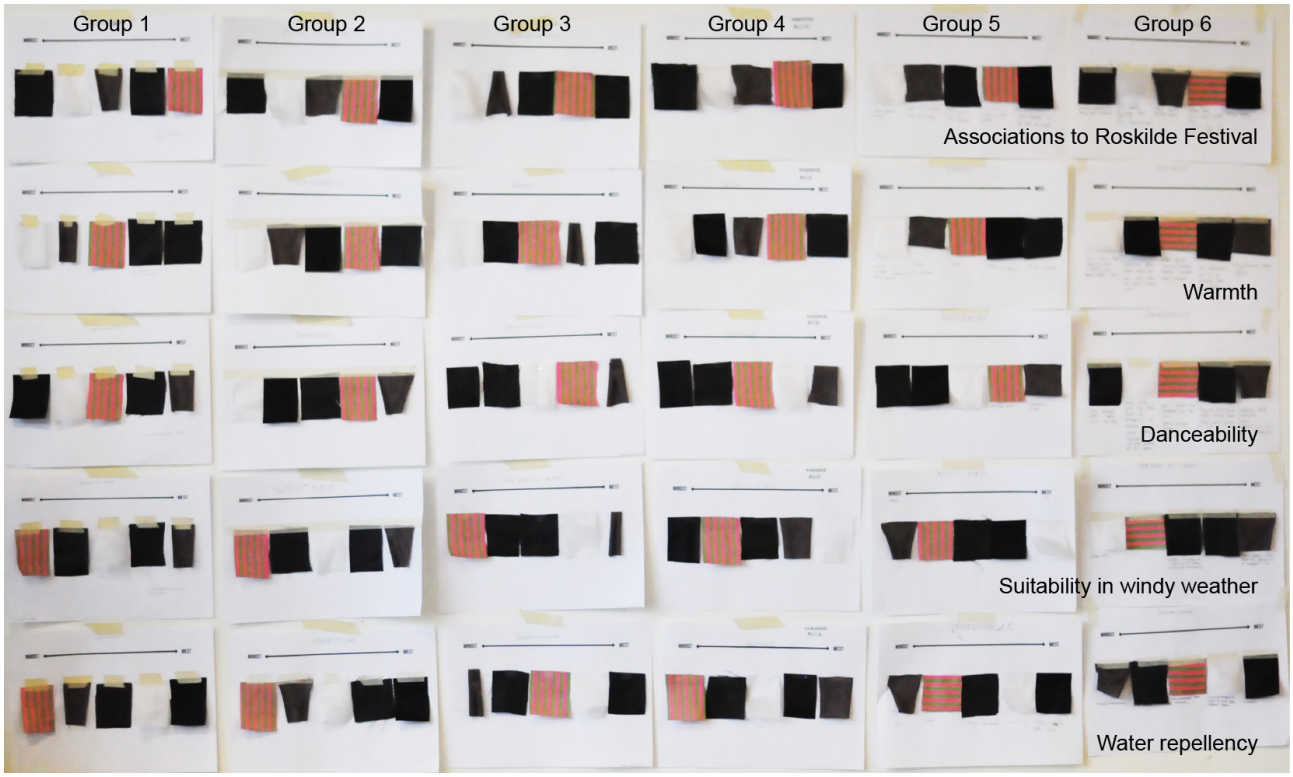


Figure 4. Examples of comparative scales using the attributes: 'associations to Roskilde Festival', 'warmth', 'danceability', 'suitability in windy weather' and 'water repellency' to order five textile materials in a 2nd semester materials introductory course for fashion and design students at Design School Kolding.

semester materials introductory course for fashion and design students at Design School Kolding. In the exercise groups of students were asked to order five materials with respect to five attributes being 'associations to Roskilde Festival', 'warmth', 'danceability', 'suitability in windy weather' and 'water repellency'. In the matrix, horizontally aligned scales share the same attribute, while vertically aligned scales are made by the same group. After the exercise, students evaluated and discussed, why the order of material samples for some attributes are more agreed on based on objective and subjective attributes. This regularly leads to a further discussion on the role of context and experience when evaluating different kinds of attributes.

While the previous exercises are graphical and verbal, translating emotional and/or associative attributes into physical materials or material compositions, an 'associative meaning translation' exercise creates strong links between tangible and intangible mental constructs and thus between different ways to create meanings of materials (Hasling, 2015; Hasling and Bang, in press). From experience with this kind of exercise it is evident that students commonly translate into similar physical means such as materials and manufacturing techniques and that they put strong emphasis on visual appearance. In Figure 5, six examples of material samples of associative meaning translations from the key phase 'Drizzle' to physical material samples from a 2nd semester materials introduction course for fashion and textiles students at Design School Kolding are shown.

Model 2: Navigating within dimensions of sustainable product design

Bridging physical and experiential understandings of materials is a first step to acknowledging the role and influence of experiential values in product design. In sustainable design practice similar approaches can be taken to appreciate different dimensions and further bridge objective and subjective understandings of, what sustainable product design can be. In the introduction it was written that sustainable development faces an increasing level of complexity, which means that designers can take multiple different approaches to sustainable design. A strong interest in materials does not mean that everything should center on developing new materials and using less resource consuming materials, but to understand, how materials use in sustainable design concepts influences and may support the concepts' overall scopes.

In the past sustainable design has been much influenced by technical and economic interests, which for example can be demonstrated by the triple bottom line approach (Elkington, 1997; Fisk, 2010). It is increasingly recognized that sustainable initiatives, or 'articulations of sustainable technology' influence each other and take part of larger networks where initiatives positively and/or negatively influence each other (Ashby & Johnson, 2014; Mulder, Ferrer & Van Lente, 2011). From courses in materials and sustainability taught in artistic design education it is however evident that students find it difficult to relate to this trinity and that something is missing in order for students (and practitioners) to take fully part of



Figure 5. Examples of associative meaning translations from the key phase 'Drizzle' to physical material samples in a 2nd semester materials introduction course for fashion and textiles students at Design School Kolding.

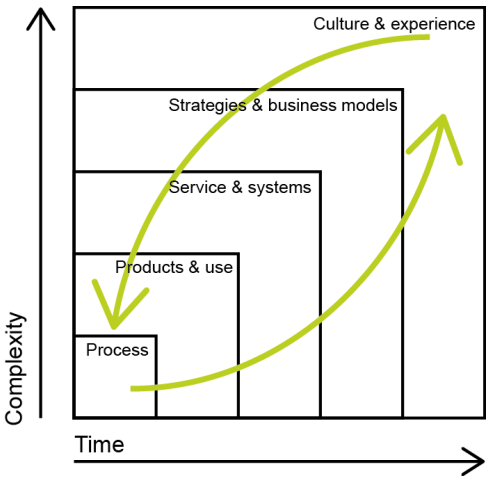


Figure 6. The five perspectives model linking 'raw materials and processes, 'product and use', 'service and system', 'strategy and business models' and 'culture and experience' of working with materials in sustainable design practice.

the sustainability agenda. Consequently, based on similar observations, Fleming and Sherman have proposed a Quadruple Bottom Line that incorporates an extra pillar being 'experience' (Fleming, 2013, 2014).

The presented five perspectives model has been developed as a way to navigate within the realm of sustainable design on the basis of conversations with product design students. Wanting to and being expected to deal with sustainability issues in their projects, students found it difficult to grasp its different perspectives and levels without leaving out essential details. Several students for example found it challenging both to go into detail with sustainable design as promoting environmental friendly material processes concurrently with considering and integrating larger systems such as dealing with services, strategies and business models in projects. They felt they had to choose their stance, which made it relevant to elaborate on, how and where materials can be positioned in the seemingly ever-developing field of sustainable development.

The five perspectives model

The five perspectives model comprises the above tendencies and streams within sustainable design and has found inspiration in notions and concepts used by for example Keitsch (2015), Bhamra and Lofthouse (2007), Vezzoli and Manzini (2010) and McLennan (2004) and with an *intention-consequence* relationship inspired by Mulder et al. (2011). The five perspectives being: 'raw materials and processes', 'products and use', 'services and systems', 'strategies and business models' and 'culture and experiences' are hierarchically structured in a shell like box arrangement as illustrated in Figure 6. The arrangement illustrates, how perspectives build on and supplement each other in the depth of working with sustainable design.

In the following the five perspectives will be briefly described.

— *The 'raw materials and processes' perspective refers to the environmental impacts of extracting and producing raw materials and manufacturing products in pre-consumption and waste generation processes in post-consumption phases and is predominantly assessed by quantitative measures such as water, energy and chemical consumption as well as waste generation.*
— *The 'products and use' perspective broadens the scope of the previous perspective to include environmental impacts of consumption in a 'cradle-to-grave' or lifecycle approach adding resources consumed for transportation and while using products.*
— *The 'services and systems' perspective shifts focus from single products and individual consumers to larger systems of material and product flows that can be interchanged and reused between multiple actors. Still with focus on environmental impacts, the perspective introduces ways to share products and streamline consumption.*
— *The 'strategies and business models' perspective embraces the three inner perspectives in a holistic understanding, emphasizing economic, environmental and social sustainability in strategic design and corporate business models. This perspective also builds on a 'circular economy', where circular material flows can be used to establish a restorative economy.*
— *The 'culture and experience' perspective emphasizes experience as a means to create value with initiatives that strengthen the consumer-product relationship or that prolongs the lifetime with having multiple consumers.*

In the model the perspectives are illustrated as rather partitioned, but in reality many initiatives overlap and incorporate several concepts from different perspectives, which highlight that initiatives can be simultaneously subjectively and objectively oriented. It also highlights that more complex aspects such as sustainable business models or cultural interventions often include less complex aspects such as raw material extraction (raw materials and processes) and consumption (products and use). As a business model eco-design typically relates to raw materials and

processes and product and use, but its value is created, because users are attracted to the idea of more sustainable products, which is an experience and cultural dependent. Similarly social innovation emphasizes experience and cultural practices (Manzini, 2015), but the experiential potential is created in the interplay between all five perspectives.

Products with holistic sustainable design approaches
To elaborate on the hierarchy and cross-hierarchical interaction, the Fairphone, a phone sold as a 'fair' phone, can be used as a case. At first sight Fairphone seems like any other electronic product, but the five perspectives model can be used to illuminate, how it stands out with respect to sustainable design. Figure 7 shows the Fairphone and its parts and Figure 8 shows, how aspects of the Fairphone as a company and product can be positioned using the five perspectives model.

In the following the five perspectives model is used to elaborate on some of the aspects of the Fairphone as a sustainable product design concept.

— **Raw materials and processes.** *The company predominantly uses conflict free minerals for the phone's electronic parts. However they also admit that for all parts it is simply not possible to source all materials according to the company's ethical standards.*
— **Product and use.** *The phone's materials can be taken apart and parts can be disassembled using a conventional screwdriver, improving the chances of recycling parts. The phone's battery can be removed, making it possible to change to a newer battery or switch between multiple batteries if necessary. The phone has room for two SIM cards reducing the need for multiple phones and in addition to its build in 16GB storage, an external SD-card can be used to for storage.*
— **Service and system.** *Fairphone has no physical stores, but a well-built online platform with detailed information and where users can get instant help from other users or the Dutch service team. Fairphone uses an Android OS specially developed for the phone, but it can be customized for special wishes or be replaced by other operating systems. Fairphone offers open source 3D printing files so users can customize and manufacture their own cases.*
— **Strategy and business model.** *The business model builds on transparency and user involvement. The price calculation for the phone is transparent and similar to crowd funding webpages such as Kickstarter and Indigogo, users sign up and pay for phones before new batches are put into production. For the company it ensures that production is economically sustainable and for users it gives a kind of ownership. Part of the money goes to supporting the Chinese laborers' right to be members of labor organizations.*
— **Culture and experience.** *The Fairphone is not the best phone on the market, but the community you become part of as a user makes you feel part of something that wants to do things differently. On their blog and ongoing newsletters, users are included in concurrent developments on everything from fair raw material extraction, troubles with Chinese authorities and malfunctions in previous batches. Users are also invited to take part of the Fairphone experience, when the team visits mining areas in Africa and the assembly lines in China, which makes you feel part of a bigger, but still graspable system.*



Figure 7. Parts of the Fairphone (1st edition, 2nd batch from 2014) (source: author's own phone).

Making sustainable product design graspable in materials teaching
As demonstrated with the Fairphone, the model can help making sustainable product design easier to grasp and navigate in. It also provides a way to incorporate objective and subjective aspects and make them interact. The model can be used as a mental foundation for exercises introduced in materials courses with emphasis on sustainable design to make concepts more concrete and to position projects within a sustainable agenda. However in order to navigate between different modes of sustainability, students need to get insights and practice ways to deal with, discuss and work with sustainability such as in exercises with focus on single aspects or based on their own practical work. Here exercises can help mature students' cognitive comprehension and awareness and enable students to work independently, open-minded and holistically with sustainability as a design frame rather than an obstacle.

In the following three exercises are briefly outlined to exemplify approaches to working with sustainable product design in education. In teaching the exercises are parts of a methodology, so even though they here appear as single units, in reality they are preceded and succeeded by other activities. The three exercises have been picked out, as they demonstrate different levels of cognitive learning approaches. Here the first exercise focuses on understanding, the second exercise focuses on applying and analyzing, and the third exercise focuses on evaluating and creating.

The first exercise is an 'identification of sustainability approach' exercise, where students are trained in understanding incentives for developing and manufacturing existing products and to become aware of some of the arguments companies use to sell their products as 'sustainable'. The exercise can also be

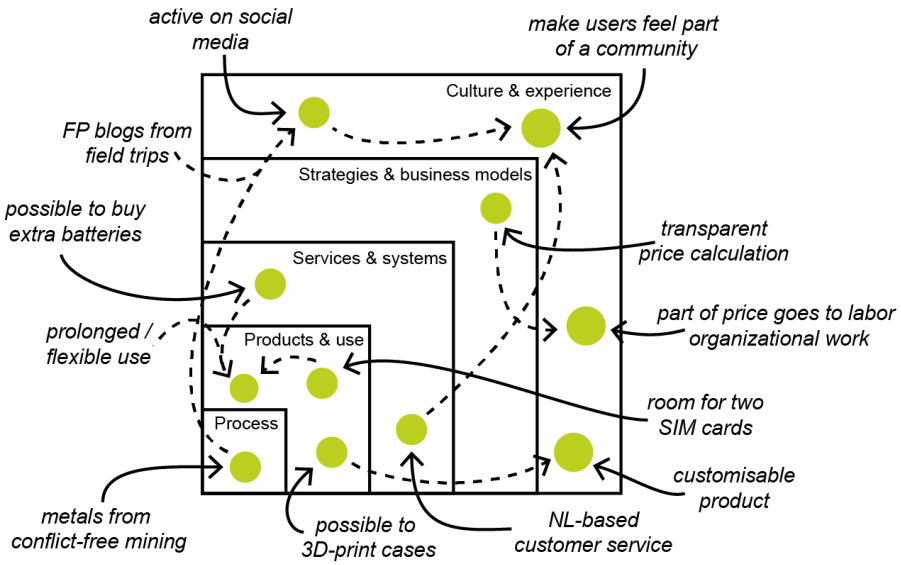


Figure 8. The Fairphone used as a case to demonstrate the five perspectives model. Here the green dots indicate specific sustainability-oriented aspects of the company / product and the striped lines indicate, how the aspects affect each other.

used as a way to make students aware of the plethora of approaches to sustainable design an, how products marketed as sustainable are distributed in the five perspectives model.

The second exercise has been inspired by Mulder et al.'s articulations of sustainable technology (Mulder et al., 2011) and their networks of articulations/ intentions-consequences. Here students are asked to graphically sketch, how aspects may affect other aspects positively and/or negatively in a 'sustainable approaches network' exercise. For inspiration, students can use 'sustainability lab cards', a collection of fifty methods cards to work with sustainable product design (Laboratory for Sustainability, 2013). Figure 9 demonstrates, how the 'sustainability lab cards' can be used. The exercise trains students in critically overviewing interactions between sustainable design approaches and reduces the risk of presenting less contemplated concepts. It further helps to raise awareness of the fact that sustainable intentions can become unsustainable but also that intentions that are not consciously sustainable can become sustainable in the right contexts.

The third exercise is a larger assignment, where students are asked to choose a minimum of three approaches, for example from from the 'sustainability lab cards' collection and develop concepts that embraces all of them. The approaches provide a frame to explore the potentials of sustainable design and students are encouraged to use their experience as practitioners to challenge conventional ways to approach sustainable design and to explore, how approaches to sustainability can work together or collide.

Discussion

The two introduced models elaborate on the same issue, namely translating meanings and shifting focus back and forth between objective and subjective understandings of the world and thus how it should be approached. While the 'physical-experiential scale of material attributes' model helps students reflect on, how to assess, communicate and choose physical materials, the 'five perspectives' model helps students to make sustainable design practice graspable and to become more eager to challenge their own and others' understanding of, what sustainable design can be.

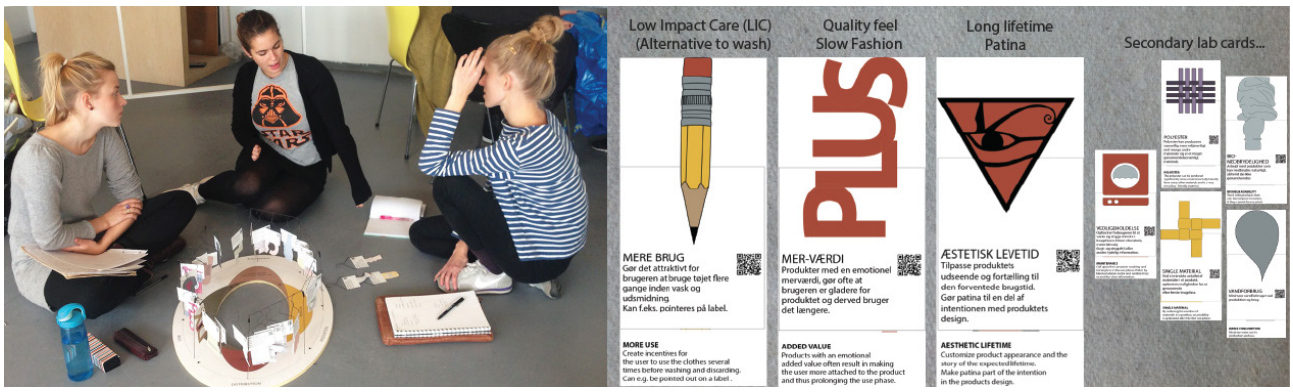


Figure 9. Students getting acquainted with the 'Lab Cards' (left), examples of lab cards selected by a group of students being 'more use', 'more value' and 'aesthetic longevity' (right) in two materials and sustainability courses for fashion and textiles design at Design School Kolding.

Together they offer a holistic approach to teaching materials that increasingly train students in navigating between different disciplines and on different cognitive levels using notions and concepts used within design practice. The models can be used as means to link materials to notions and concepts such as user experience, product experience, co-creation, participatory design, social innovation etc. within a sustainable design agenda. As the field of sustainable development increases in size and complexity, the models can be used to create common understandings across disciplinary boundaries.

Conclusion

The paper elaborates on materials teaching in design education emphasizing a higher degree of appreciation of experiential aspects of materials in fundamental materials courses as well as in courses on sustainable product design. This emphasis should create a stronger interaction between physical and experiential values to consider materials in sustainable design education. Based on two models, a model that presents a physical-experiential scale of material attributes and a model that presents approaches to sustainable design as five hierarchical perspectives, six practical exercises are introduced. The exercises serve to prepare students to increasingly work with experiential attributes in sustainable design education and thus relating it to their own practice as designers.

The models have been introduced together to emphasize the importance of integrating an experiential way of thinking early on in the curriculum and to show that working with sustainable aspects can be regarded as an additional dimension to existing curricular activities rather than being something different.

The paper suggests that with the exercises and elaborate focus on the diverse ways of understanding and communicating attributes in materials and products, students become more aware of the range of and correlation between different kinds of attributes in understanding materials as well as sustainable product design. Furthermore, because experiential values are emphasized, students find it easier also to relate more technical aspects to their own practice. It can be stated that interaction between physical and experiential values is a key to understanding materials as well as sustainable design practice and thus experiential dimensions can and should increasingly be regarded as co-drivers for sustainable design.

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